Novel mathematical insights from the proof-refinement loop (Lemma 5.4 of Braverman et al., *The Grothendieck constant is strictly smaller than Krivine's bound*)

Appendix to the main paper. This appendix collects the genuine mathematical content discovered while driving the constant C_r in the rigorous certificate of Lemma 5.4 down through several orders of magnitude. We follow the convention that $\mathcal{K}_G$ denotes the (real) Grothendieck constant. The Krivine upper bound is $\mathcal{K}_G \le \pi/(2\ln(1+\sqrt{2})) \approx 1.7822$; each refinement of the inequality below produces a strictly smaller certified upper bound for $\mathcal{K}_G$. Throughout, $\eta = 1/100$ and $r = 23/25$ unless stated otherwise, and $\Omega(\delta, B) \subset \mathbb{C}$ denotes the rectangle $\{w : |\Re w| \le 1-\delta,\ |\Im w| \le B\}$ on which the implicit-function family is analytic.

1 The improvement-over-Krivine quantity

In the Braverman–Guruswami–Lee–Naor program, the certified upper bound on $\mathcal{K}_G$ takes the form

$$\mathcal{K}_G \ \le\ \frac{1}{c_{f,g}}, \qquad c_{f,g} = \tfrac{2}{\pi}\,\gamma, \qquad \gamma = \gamma_0 + \Delta\gamma, \tag{1}$$

where $\gamma_0 = \operatorname{atanh}(1/\sqrt{2}) = \ln(1+\sqrt{2})$ is the value the inequality collapses to in the absence of the Lemma 5.4 refinement. Setting $\Delta\gamma = 0$ recovers the classical Krivine bound $\pi/(2\ln(1+\sqrt{2}))$. The whole program is the construction of a strictly positive, rigorously certified $\Delta\gamma > 0$. The natural *quantity of merit* is therefore the gap between Krivine's constant and the new upper bound,

$$\mathcal{K}_G\text{-improvement} \ :=\ \frac{\pi}{2\ln(1+\sqrt{2})} \ -\ \frac{1}{c_{f,g}} \ =\ \frac{\Delta\mathcal{K}_G}{c_{\text{base}}\,(c_{\text{base}} + \Delta\mathcal{K}_G)}, \tag{2}$$

with $c_{\text{base}} = (2/\pi)\gamma_0$ and $\Delta\mathcal{K}_G = (2/\pi)\Delta\gamma$. Two properties of (2) are used systematically in the audits and figures.

Inversion stability. Because $c_{\text{base}} = (2/\pi)\ln(1+\sqrt{2}) \approx 0.5611$ is bounded away from zero, the radius of the interval $1/c_{f,g}$ in arb (interval-ball) arithmetic is comparable to the radius of $c_{f,g}$ itself: a relative widening by the factor $1/c_{\text{base}}^2 \approx 3.18$. Subtraction with the constant $\pi/(2\gamma_0)$ does *not* suffer cancellation in arb because the radius is bounded above by the upstream stored radius $\sim 10^{-41}$, well below the actual gap $\Delta\mathcal{K}_G/c_{\text{base}}^2$ which lies between 10^{-32} and 10^{-17} across iterates. This guarantees that the plotted improvement values are not artifacts of floating-point cancellation.

Linearisation. For $\Delta\mathcal{K}_G \ll c_{\text{base}}$, (2) expands as $\mathcal{K}_G$-improvement $= (1/c_{\text{base}}^2)\,\Delta\mathcal{K}_G + O(\Delta\mathcal{K}_G^2) = \big(\pi/(2\gamma_0)\big)^2\Delta\mathcal{K}_G + O(\Delta\mathcal{K}_G^2)$, so the curve $\mathcal{K}_G$-improvement vs. iteration is, to leading order, just the Krivine constant squared times the $\Delta\mathcal{K}_G$ trace. The plots in `plots/kg_improvement.pdf` display the exact form (2) computed with arb at precision 256.

2 The auxiliary z-identity for $\|u_{zz}\|$

The original argument bounds $\|u_{zz}\|$ via a recentred Cauchy estimate on the analyticity disk $\{|z - z_0| < r\}$: $\|u_{zz}\| \le 8R/(1-r)^2$. The new derivation comes from differentiating the implicit-function identity $F(\eta, p, F^{-1}(p,z)) = z$ twice in z and exploiting the algebraic relation

$$D \cdot u_{zz} \;+\; DD \cdot h_z^2 \;=\; 0, \qquad D = \partial_w F, \quad DD = \partial_w^2 F, \quad h_z = \partial_z F^{-1} = 1/D. \tag{3}$$

Solving (3) gives the *exact pointwise identity*

$$u_{zz} \;=\; -\,\frac{DD\, h_z^2}{D} \;=\; -\,\frac{DD}{D^3}. \tag{4}$$

The auxiliary z-identity (3) is, up to sign, a re-statement of $\partial_z(Dh_z) = 0$ along the inverse curve. Identity (4) replaces the worst-case Cauchy step $8R/(1-r)^2 \approx 1393$ by a direct sup $\|u_{zz}\| \le \left\|DD/D^3\right\|_\Omega$ which our truncation-free chunk certifier evaluates to a value ≈ 1.55, a $\sim 900\times$ tightening of the term that dominated the V10 baseline. Crucially this is identity (not inequality) at every (p, w), so the slack is purely a function of how tightly we can sup over a given (p, w)-cell.

3 Joint sup-bounds vs. triangle products

A recurring leverage point is that the Lemma 5.4 derivation produces inequalities of the form

$$\sup_{(p,w)} \left\| A(p,w)\, B(p,w) / \; C(p,w)^k \right\| \;\le\; \Big(\sup A\Big)\,\Big(\sup B\Big) / \Big(\inf C\Big)^k, \tag{5}$$

and the right-hand side is what the original proof bounds by *three independent* suprema. (5) is sharp only when the three suprema all bind at the same (p, w). For our analytic family they bind at *different* corners, and the right-hand side over-estimates the joint sup by a factor that turns out to be 5–300$\times$ depending on the cell.

Term 1 (sprint 019). The triangle product $M_\phi\, M_{\phi'}\, (1/D)^2$ on the V13 oracle constellation evaluates to 2.09×10^{-3}, while the joint sup $J_1 \;=\; \sup_{(p,w)} \left\|\Phi(w)\, \Phi'(w) \,/\, D(p,w)^2\right\|$ certifies to $J_1 \le 3 \times 10^{-4}$ on the same rectangle—a $\sim 7\times$ reduction. Mathematically this happens because $\Phi(w)$ and $\Phi'(w)$ are η-*correlated*: each is an $O(\eta^2)$ deviation from $H_0(w)$ resp. $H_0'(w)$, and the deviations have aligned phase, so $|\Phi\, \Phi'|$ at the binding w is much smaller than the product of the worst-case magnitudes.

Term 2 (sprint 020). The same mechanism applied to the second term: $\frac{1}{2} u_{zz}\, M_\phi^2 = \frac{1}{2}\, \left\|DD/D^3\right\|\, M_\phi^2 \approx 9.24 \times 10^{-4}$ becomes $J_2 \;=\; \frac{1}{2} \sup_{(p,w)} \left\|DD(p,w)\, \Phi(w)^2 / D(p,w)^3\right\| \approx 7.7 \times 10^{-5}$, a $\sim 6\times$ reduction. The *same* aux-z identity $u_{zz} = -DD/D^3$ explains why $u_{zz}\, \Phi^2 = -DD\, \Phi^2/D^3$ is the natural single-cell quantity to certify.

4 The form-A identity and merged-numerator interference

Form-A identity. Differentiating $F(\eta, p, F^{-1}(p,z)) = z$ twice in p at fixed z yields

$$D\,u_{pp} \;+\; 2\,\Phi'\,u_p \;+\; DD\,u_p^2 \;=\; 0, \tag{6}$$

where $u_p = \partial_p F^{-1}$ satisfies, by a separate differentiation, $u_p = -\Phi/D$ with $\Phi = H_\eta - H_0$. Substituting $u_p = -\Phi/D$ into (6) and solving gives the *exact pointwise identity*

$$u_{pp} \;=\; \frac{2\,\Phi(w)\,\Phi'(w)\,D(p,w) \;-\; DD(p,w)\,\Phi(w)^2}{D(p,w)^3}. \tag{7}$$

This is the *merged numerator*. The two terms in the numerator have opposite signs and partially cancel.

Triangle vs. merged. A naive triangle bound on $|u_{pp}|$ would split (7) into $|u_{pp}| \;\le\; 2\,\big\|\Phi\Phi'/D^2\big\| + \big\|DD\,\Phi^2/D^3\big\| \;=\; 2J_1 + J_2$, discarding the destructive interference between the two summands. The merged sup $J_{\text{merged}} \;=\; \sup_{(p,w)} \frac{|2\Phi\Phi' D - DD\,\Phi^2|}{|D|^3}$ is the genuine pointwise sup of $|u_{pp}|$, and $J_{\text{merged}} \le 2J_1 + J_2$ with equality if the two summands are everywhere phase-aligned.

Resolution-dependent crossover. A surprising phenomenon emerges when computing J_{merged} rigorously by interval arithmetic on cells of $\partial\Omega \times [0,1]$ at resolution n:

n	1,024	2,048	4,096	8,192	16,384	32,768	65,536	131,072
$J_{\text{merged}}/(2J_1+J_2)$	1.99	1.92	1.87	1.78	1.61	1.40	1.20	1.034

At low resolution the certified bound on the *difference* $|2\Phi\Phi' D - DD\Phi^2|$ is dominated by the per-cell interval slack of subtracting two large quantities of opposite sign—the bound is *worse* than the triangle. The cancellation is real but invisible at coarse scale. Around $n \approx 2^{17} - 2^{18}$ the per-cell slack shrinks below the cancellation gain and the merged bound crosses below the triangle. At $n = 262{,}144$ the certified merged bound is $\approx 0.86\times$ the triangle, yielding a structural $\sim 14\%$ reduction that the triangle decomposition cannot capture at any resolution.

This crossover is the cleanest example we have observed of a rigorous bound that is strictly tighter only above a quantitative resolution threshold, where "resolution" refers to the discretisation of the sup.

5 The $K_{\max}$ truncation regression and a truncation-free pipeline

The semi-analytic step in the original numerical pipeline writes

$$H_\eta(w) \;=\; \arcsin(w) \;-\; \frac{\phi\big(\arcsin(w)\big)}{\sqrt{1-w^2}}, \qquad \phi(z) \;=\; \sum_{k\ge 0} c_k\, z^k, \tag{8}$$

and evaluates ϕ by truncating the series at $K_{\max}$. The b-series file at our parameters has 250 coefficients with empirical pseudo-rate $|b_k|^{1/k} \approx 0.964$ near $k = 249$; a truncation at $K_{\max} = 80$ leaves a tail that is *not bounded rigorously* for the magnitudes that arise in $\Phi = H_\eta - H_0$ at $\eta = 1/100$ (Φ itself is $O(\eta^2) \sim 10^{-4}$, whereas the pseudo-rigorous tail is plausibly $O(10^{-8})$ but unverified). Interval arithmetic over a truncated polynomial returns a rigorous bound on the wrong function.

5.1 Integral representation via Price's theorem

The replacement is to evaluate $H_\eta, H'_\eta, H''_\eta$ *without* ϕ. Following Price's theorem, the analytic continuation of H_η to $w \in \mathbb{C}$ admits the integral representation

$$H_\eta(w) \;=\; \frac{1}{2\sigma_z} \iint_{\mathbb{R}^2} K_z\big(P(x), Q(y)\big)\, e^{-S_z[P,Q](x,y)}\, dx\, dy, \tag{9}$$

where $K_z(P,Q) = \operatorname{sgn}(x - \eta P(x)) \operatorname{sgn}(y + \eta Q(y))$ for the Hermite-physicist sign-pair we use, S_z is the action quadratic form parameterised by $z \in \Omega$, and σ_z is the corresponding normalising determinant. The first and second w-derivatives are obtained by differentiating (9) under the integral, producing analogous formulas for H'_η and H''_η that involve P', Q' but no power series in w.

The integrals (9) are evaluated via tensor-product Gauss–Legendre at the chunk midpoint, with adaptive Bernstein-ellipse error per panel; the value is then a sharp rigorous evaluation of $H_\eta(w)$ (and its derivatives) at that midpoint.

5.2 Real-positive majorants for chunk bounds

To bound $\sup_{w \in Z} |H_\eta(w)|$ over a chunk $Z \subset \Omega$, we replace (9) with its real-positive majorant

$$|H_\eta(w)| \;\le\; \frac{1}{2|\sigma_z|} \iint_{\mathbb{R}^2} \big|K_z(P,Q)\big|\, \big|e^{-S_z}\big|\, dx\, dy \;\le\; \frac{1}{2\underline{|\sigma_z|}} \iint_{\mathbb{R}^2} \widehat{E}_Z(x,y)\, dx\, dy, \tag{10}$$

where $\widehat{E}_Z(x,y)$ is a real, non-negative Gaussian-shaped majorant of $|K_z\, e^{-S_z}|$ valid uniformly for $w \in Z$. The right-hand side reduces to a finite sum of closed-form Gaussian moments $\int x^{2j} e^{-\alpha x^2}\, dx$ that arb evaluates rigorously. The chunk-bound for $H'_\eta, H''_\eta, \Phi, \Phi'$ are analogous, with a key extra ingredient described below.

5.3 The η^2-tight Φ-difference bound

Bounding $\Phi(w) = H_\eta(w) - H_0(w)$ by a triangle inequality $|\Phi| \le |H_\eta| + |H_0| \approx 2.4$ throws away the $O(\eta^2)$ scaling of Φ and produces a C_r that is six orders of magnitude worse than even the heuristic prior baseline. The remedy is to take the difference *inside* the integral:

$$\Phi(w) \;=\; \frac{1}{2\sigma_z} \iint_{\mathbb{R}^2} \big[K_z(P,Q) - K_z(0,0)\big]\, e^{-S_z[P,Q]}\, dx\, dy \;+\; \text{(small) Gaussian-action correction,} \tag{11}$$

and majorise the kernel difference $K_z(P,Q) - K_z(0,0)$ by a real-positive Taylor expansion in (P,Q) around $(0,0)$. Because P, Q are themselves $O(\eta)$, the chunk-bound on Φ scales as $O(\eta^2)$, recovering the cancellation that the triangle bound discarded.

5.4 The λ-decay parallelogram fix

A subtle bug in an earlier version of the H''_η chunk certifier used the bound $\lambda \le 2\min(\alpha, \beta)$, where α, β are the principal-axis decay rates of the Gaussian factor. The correct bound is

$$\lambda \;\le\; \min(\alpha, \beta), \tag{12}$$

which is the parallelogram identity $(x-y)^2 + (x+y)^2 = 2(x^2+y^2)$ in disguise: the joint quadratic form on the *difference* of two Gaussians has decay $\min(\alpha, \beta)$, not $2\min(\alpha, \beta)$. The factor-of-two error inflated the certified Gaussian moments by 2^j at the $2j$-th moment and was the main quantitative gap between the truncation-free pipeline and the heuristic baseline before the fix.

5.5 Anisotropic Gaussian moments

Every chunk-bound integral collapses to a sum of moments of an anisotropic Gaussian $e^{-(\alpha x^2+\beta y^2+2\gamma xy)}$. Bounding each moment by its isotropic majorant (replace α, β by $\min(\alpha, \beta)$) is sound but loose. We use moment $\le \min\big(\text{isotropic-majorant},\ \text{exact-anisotropic}\big)$, where the exact-anisotropic value is computed by diagonalising the quadratic form via an explicit rotation. The minimum is provably an upper bound on the moment and is tighter than either alone on cells where α, β differ substantially.

6 The denominator-fix for $D(p, w)$

The joint quotients $\Phi\Phi'/D^2$ and $DD\,\Phi^2/D^3$ require a uniform lower bound $|D(p, w)| \ge \delta_D > 0$ on each cell. Computing $|D|$ directly as a joint (p, w) interval inflates by orders of magnitude because D is a difference $D = H'_0 + p(H'_\eta - H'_0)$ where the two summands have similar magnitude. The fix exploits the affine structure in p:

$$|D(p, w)| \ \ge\ |H'_0(w)| \ -\ |p| \cdot |H'_\eta(w) - H'_0(w)| \ \ge\ \underline{|H'_0(w)|} \ -\ p_{\text{hi}} \cdot M_{\Phi'}, \tag{13}$$

where $M_{\Phi'}$ is the chunk-certified bound on $|\Phi'|$ and $\underline{|H'_0(w)|}$ is the closed-form interval lower bound $1/|1 - w^2|^{1/2}$. The right-hand side of (13) is computed once per w-chunk (independent of p) and is positive provided the chunk is bounded away from the branch points $w = \pm 1$. This avoids multiplying a small p-interval by a large $H'_\eta - H'_0$ in the joint (p, w) representation and recovers near-optimal denominator bounds at every cell.

7 The 2nd-order Taylor pad in w

A chunk Z of w-radius r_Z admits a sharp interior midpoint w_0 at which $H_\eta, H'_\eta, H''_\eta$ are evaluated by (9). The chunk values are then related to the midpoint values by 2nd-order Taylor expansion in w:

$$\begin{aligned}
\sup_{w\in Z} |\Phi(w)| &\le |\Phi(w_0)| + \tfrac{r_Z^2}{2}\, M_{\Phi''} + \tfrac{r_Z^3}{6}\, M_{\Phi'''},\\
\sup_{w\in Z} |\Phi'(w)| &\le |\Phi'(w_0)| + r_Z\, M_{\Phi''} + \tfrac{r_Z^2}{2}\, M_{\Phi'''},\\
\sup_{w\in Z} |\Phi''(w)| &\le \min\big(|\Phi''(w_0)| + r_Z\, M_{\Phi'''},\ M_{\Phi''}\big),
\end{aligned} \tag{14}$$

where $M_{\Phi'''}$ is a Cauchy bound on the third derivative obtained from the analyticity of Φ on a slightly larger disk. The padded values are sharper than either the chunk-uniform bound $M_\bullet$ or the midpoint value alone, and are the mechanism by which corner-subdivision rounds (cf. §9) drive the certified sup down: each subdivision halves r_Z and the pads shrink at order r_Z^2, r_Z, and 1 respectively, with the third estimate switching from the Cauchy-padded value to the chunk-uniform bound at small r_Z.

8 Per-cell $\min(\textbf{interval}, \textbf{norm})$

For cells where the merged numerator $2\Phi\Phi' D - DD\,\Phi^2$ has matched-magnitude summands of opposite sign, the interval-arithmetic numerator $\text{score}_{\text{interval}} = \overline{|2\Phi\Phi' D - DD\Phi^2|}/\underline{|D|}^3$ captures the destructive interference but suffers dependency-blow-up when the constituents widen across the cell. The triangle-norm form $\text{score}_{\text{norm}} = 2\,\overline{|\Phi||\Phi'|}/\underline{|D|}^2 + \overline{|DD||\Phi|}^2/\underline{|D|}^3$ discards interference but has no dependency-blow-up. Both quantities are valid upper bounds on $|u_{pp}|$ on the cell. We always use the tighter of the two, $\text{score}_{\text{cell}} = \min(\text{score}_{\text{interval}}, \text{score}_{\text{norm}})$, which is again provably a valid upper bound. The trade-off shifts cell-by-cell: near corner saturation, $\text{score}_{\text{norm}}$ wins; in the bulk, $\text{score}_{\text{interval}}$ wins.

9 Symmetry of the V26 contour and corner subdivision

Four-fold symmetry. The function H_η at our parameters satisfies the two reflection identities $H_\eta(\bar{z}) = \overline{H_\eta(z)}$, $H_\eta(-z) = -H_\eta(z)$. The first is a consequence of the real-coefficient Taylor expansion of H_η near the origin; the second is the Hermite-physicist parity of the sign-pair K_z. The joint four-fold symmetry reduces the chunk certifier's work by exactly $4\times$: only w-cells in the first quadrant of Ω need to be evaluated primarily.

Corner subdivision. The chunk certifier uses a uniform $n_\bullet \times 4$ partition of $\partial\Omega$, but the sup is consistently dominated by chunks adjacent to the four corners $\pm(1-\delta) \pm iB$. Recursively splitting each corner chunk 4-way (or 2-way for 1D edge chunks) and re-running the Taylor-pad evaluator on the sub-chunks halves the certified sup roughly each round:

round	1	2	3
C_r	2.27	1.23	0.383

starting from $C_r = 34.47$ at the unsubdivided 64-chunk baseline. The diminishing returns reflect the fact that the binding cells move inwards from the corners as the local bound becomes tighter than the bulk.

10 The C_r landscape: formula $\times$ oracle constellation

Each candidate V_K in the refinement loop encodes a pair (C_r formula, oracle constellation), not a linear sequence of literals. After the truncation-free re-audit (§5), the *patched* C_r values rank as follows for our target parameters $\eta = 1/100$, $r = 23/25$, $p_{\max} = 1$:

V_K	C_r formula	C_r^{patched}
V_{11}	$C_r^{\phi,u_{zz}} = M_\phi M_{\phi'} I^2 + \frac{1}{2}\,\|u_{zz}\|\,M_\phi^2$	0.3255
V_{14}	$C_r^{\phi,u_{zz},\text{joint}} = J_1 + \frac{1}{2}\,\|u_{zz}\|\,M_\phi^2$	**0.2810**
V_{15-24}	$C_r^{\phi,u_{zz},\text{joint2}} = J_1 + \frac{1}{2}J_2$	0.4105
$V_{25,26}$	$C_r^{\phi,u_{zz},\text{merged}} = \frac{1}{2}\,J_{\text{merged}}$	0.3734

($I = \sup 1/|D|$.) The smallest C_r^{patched} is held by V_{14}. Two surprises emerge.

Joint-on-only-one-term. V_{14}'s mixed formula—J_1 (joint sup, term 1) plus $\|u_{zz}\|\, M_\phi^2$ (triangle product, term 2)—wins over $V_{15\text{-}24}$'s $J_1 + \frac{1}{2}J_2$ (joint sup on *both* terms). This is a consequence of the form-A identity (7): replacing $\|u_{zz}\|\, M_\phi^2$ by $J_2 = \sup \left\| DD\, \Phi^2 / D^3 \right\|$ changes which factor (the numerator $|DD|$ or the correlation $|\Phi|^2$) is taken on its worst cell, and at our parameters the correlation is more concentrated than the joint sup over a single cell can capture, so the triangle bound on term 2 is paradoxically tighter on the patched oracles.

Merged loses (at finite resolution). $V_{25,26}$'s merged formula is exact at the level of (7), but the certified sup J_{merged} at the chunk resolutions used for the audit ($n = 2^{18}$) is still slightly above the triangle $2J_1 + J_2$, by exactly the crossover phenomenon in §4. At higher resolution ($n \geq 2^{20}$ extrapolated), the merged would dominate; at the production resolution of the audit, the joint mixture in V_{14} wins.

Take-away. Beating any specific C_r^{patched} requires *moving the audit floor*, not just tightening the literal:

1. re-baseline the formula choice onto V_{14} for an immediate $\sim 17\%$ win;
2. subdivide chunks further (each round halved J_{merged} historically);
3. replace the 2nd-order Taylor pad by a 4th-order pad with a tighter $M_{\Phi^{(4)}}$;
4. use an alternative analyticity domain (octagonal, Cassini, …) that avoids corner saturation entirely.

Items (3)–(4) are open analytic problems and require new oracles.

11 The Cauchy-radius lift and the analyticity-domain envelope

The implicit-function family F^{-1} is constructed in the original proof as the fixed-point of a contraction on a Banach space of analytic functions on $\text{disk}(r_1\, r)$ with $r_1 = (1 + 1/r)/2$. Our Lean axiomatisation supplies the strictly weaker analyticity domain $\text{disk}((1+r)/2)$ (which equals 0.96 at $r = 23/25$), forcing the recentred Cauchy ball at any $z_0 \in \text{disk}\, r$ to have radius $\rho \leq (1-r)/2 = 0.04$. By the standard Cauchy estimate $|f^{(n)}(z_0)| \leq n!\, \sup_{|z-z_0|=\rho} |f(z)|/\rho^n$, shrinking ρ inflates derivative bounds by $1/\rho^n$. Lifting the analyticity domain to $\text{disk}((3+r)/4) \approx 0.98$ enlarges the available Cauchy radius to $\rho = (3 - r)/4 \approx 0.13$, a $1.5\times$ lift, with an n-th derivative bound shrinking by 1.5^n:

- $1/D$ (first derivative): $1.5\times$ tighter,
- u_{zz} via Cauchy on the second derivative: $2.25\times$ tighter,
- third-derivative Taylor pads (§7): $3.375\times$ tighter.

The wider domain claim $\text{disk}((3 + r)/4)$ is paper-citable in the original proof's Step 4; bringing it into the Lean axiomatisation is the single largest *trust-expansion* gain available, with a downstream win factor that compounds with the chunk-bound improvements above.

12 Summary of trace

Across 34 refinement sprints driven by the framework, the certified C_r for Lemma 5.4 traces (in pre-truncation-free units, then in truncation-free units after the May-2026 patch):

$$8.2\times10^{14} \;\to\; 1.13\times10^{9} \;\to\; 3.79 \;\to\; 1 \;\to\; 1/268{,}000 \quad \xrightarrow{K_{\max}\text{ patch}} \quad 0.3255 \;\to\; 0.2857.$$

The correspondence between the C_r trace and the $\mathcal{K}_G$-improvement trace (2) is recorded in `plots/cr_progression` and `plots/kg_improvement.pdf`. The non- monotonic plateaus reflect formula switches; the steepest descents come from the joint oracles (§3) and the $K_{\max}$-truncation patch (§5).

The mathematical insights collected above—the auxiliary z-identity, the form-A identity and merged-numerator interference, the joint-vs-triangle gap, the η^2-tight Φ-difference bound, the λ-decay parallelogram fix, the denominator-fix and Taylor-pad machinery, the per-cell $\min(\text{interval}, \text{norm})$, the four-fold symmetry, and the analyticity-domain lift—are individually modest, but their composition is what makes the certified $\mathcal{K}_G$-improvement strictly positive in arb-arithmetic at the 10^{-17} scale.